14 May 84
SUPERSEDING MIL-A-26846A
(USAF)
26 Sep 1958

MILITARY SPECIFICATION

AIR CONDITIONER, A/M32C-4, TRAILER MOUNTED, VAPOR CYCLE, ELECTRIC MOTOR DRIVEN

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electric motor driven, vapor cycle, trailer mounted air conditioner, having a nominal cooling capacity of 84000 BTU/HR (7 Tons), Type A/M32C-4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

*	CC-M-1807	Motor, Alternating Current, Fractional And Integral Horsepower (500 Hp And Smaller)
	QQ-B-654	Brazing Alloy, Silver
	QQ-S-698	Steel, Sheet And Strip, Low Carbon
	VV-L-825	Lubricating Oil, Refrigerant Compressor
	WW-T-799	Tube, Copper, Seamless, Water and Refrigeration (For use with Solder - Flared - or Compression Type Fittings)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Department, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-V-173	Varnish, Moisture And Fungus Resistant (For Treatment of Communications, Electronic, And Associated Equipment)
MIL-F-3541	Fitting, Lubrication
MIL-R-3593	Refrigeration And Cooling Equipment (Excluding Household Refrigerators) Packaging Of
MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification For
MIL-C-5756	Cable and Wire, Power, Electrical, Portable
MIL-R-5757	Relay, Electrical (For Electronic and Communication Type Equipment), General Specification For
MIL-T-7928	Terminal, Lug And Splice, Crimp-style, Copper
MIL-W-8005	Wheels And Hubs, For Industrial Pneumatic Tires
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Specifications For
MIL-G-8402	Gage, Pressure, Dial Indicating General Specification For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-I-22023	Insulation Felt, thermal and Sound Absorbing Felt, Fibrous Glass, Flexible
MIL-W-22759	Wire, Electric, Fluoropolymer Insulated, Copper or copper Alloy
MIL-D-38386	Duct Assembly, Ground, Conditioned Air, Insulated, Flexible
MIL-M-43248	Mat, Reinforcing Glass Fiber
MIL-R-46068	Resin, Polyester, Bisphenol - A Type

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

DOD-STD-100	Engineering Drawing Practice
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For the Selection Of
MIL-STD-195	Marking Of Connections For Electrical Assemblies
MIL-STD-281	Automobile, Trucks, Truck-Tractors, Trailer and Trailer Dollies, Preservation and Packaging of
* MIL-STD-461	Electromagnetic Interference, Characteristics Requirement For Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

*	MIL-STD-462	Electromagnetic Interference, Characteristics Measurement Of
	MIL-STD-808	Finishes, Protective and Codes, For Finishing Schemes For Ground And Ground Support Equipment
	MIL-STD-810	Environmental Test Methods
	MIL-STD-831	Test Reports, Preparation Of
	MIL-STD-889	Metals, Dissimilar
	MIL-STD-1474	Noise Limits For Army Materiel
	MS27265	Connector, Plug, Electric Ramp Power 460/240, Male
	MS27766	Shroud For Aerospaces Ground Equipment
	MS51335	Pintle Assembly, Towing 18,000 lbs Capacity, Manual Release
	MS51336	Lunette, Coupler, Drawbar, Ring

DRAWINGS

AIR FORCE

	48B7796	Ring Assembly Tie Down 10,000 lb
	49B6564	Ring-Eyebolt, 25,000 lb Cargo Tie Down
	64C1004	Connector-Air Dut Hose, Ground Cooling Male
	64C1005	Connector-Air Duct Hose, Ground Cooling Female
*	7545352	Requirements for Finishes Protective And Codes for San Antonio ALC, Ground And Ground Support Equipment
*	8044080	Cap, Air Duct, Female, Ground Cooling
	8321218	Duct Assembly, Ground Condition, Air Insulated, Flexible

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other publications. The following documents(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF HEATING, REFRIGERATION & AIR CONDITIONING ENGINEERS

ASHRAE Incorporated Standard 37-69 - Unitary Air Conditioning Equipment, Method of Testing for Rating (Latest Edition)

AShRAE Guide and Data Book (Latest Edition)

(Application for copies may be obtained from the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., 345 E 47th Street, New York NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME Boiler and Pressure Vessel Code Section VIII Unfired Pressure Vessels (Latest Edition)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York NY 10017.)

AMERICAN STANDARDS ASSOCIATION

Standard S1.4 - 1961	General Purpose Sound Level Meters
Standard S1.6 - 1960	Preferred Frequencies for Acoustical Measurements

(Application for copies should be addressed to American Standards Association, 10 E 40th Street, New York NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70 - National Electrical Code (Electric Wiring & Apparatus)

(Application for copies should be addressed to National Fire Protection Association, 60 Batterymarch Street, Boston MA 02119.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. This specification makes provisions for First Article testing, of the air conditioner.

3.2 Components. The air conditioner shall consist of at least the following components.

Item number	Description	See requirement
1	Refrigeration system	3.7
2	Condenser assembly	3.8
3	Receiver	3.9
4	Evaporator assembly	3.10
5	Air/refrigeration controls system and accessories	3.11
6	Refrigeration piping	3.12
7	Air flow system	3.13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

8	Prime mover assembly	3.14
9	Enclosure trailer design features	3.15
10	Electrical System	3.16
11	Instruments	3.17

3.3 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as specified in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts such as screws and self-tapping screws shall not be used. Bolts, nuts and cotter pins having suitable properties shall be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exceptions specified in 3.3.1, MS and AN standard parts shall be used where they are suitable for the purpose intended. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Fungus-Proof material. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion resistant type or treated in accordance with paragraph 3.26 to resist corrosion due to fuels, salt spray or atmospheric conditions likely to be met in storage or normal service.

3.4.3. Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Gasketing and insulation materials. Where rubber or similar gasketing and insulating materials are used, they shall be compounded to insure their oil resistance and, where applicable, their ability to withstand low temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

3.5 Design and construction. The air conditioner shall be designed to be completely automatic once it is placed in operation and shall provide a rugged and trouble-free unit of lightest weight and smaller practicable size comparable with the requirements specified herein. The air conditioner shall be so designed that all components, including stored ducts and operating controls, shall be enclosed in a metallic enclosure and mounted on a 4-wheel trailer. The air conditioner shall be so designed that the major air conditioner components specified herein can be removed through enclosure doors and/or removable panels. A drawbar shall be provided in the front of the unit to provide a means of attaching a towing vehicle, also a pintle hook on the rear for towing additional vehicle. Tie down rings shall be provided to secure the unit in shipment and for lifting. The enclosure shall be so designed to present a neat, smooth external appearance free from sharp edges and sharp corners.

3.5.1 Adjustments and repairs. The air conditioner shall be so constructed that adjustments and repairs can be easily made by personnel of operating units and overhaul bases with the use of standard hand tools.

* 3.5.2 Life. The air conditioner shall be designed for an operating life of 3,000 hours between major overhauls and shall be capable of operating for 600 hours at design capacity without failure of any major part (see 4.6.11).

* 3.5.3 Forklifting provisions. The air conditioner shall be provided with lifting pads or channels along the width dimension to permit the air conditioner to be forklifted without damage. These pads or channels shall be positioned no more than 20 inches (inside dimension) on each side of the center of balance of the unit. The pads or channels shall be capable of accommodating lifting forks that are 8 inches wide and 2 3/4 inches thick.

* 3.5.4 Hoisting. The air conditioner shall be constructed with hoisting provisions to permit hoisting of the enclosure for accessibility and hoisting of the complete, dry unit (see 6.3) from the chassis. An internal spread bar shall be incorporated at the top of the enclosure and shall be designed not to collapse during hoisting. Tie down provisions in accordance with MIL-A-8421 shall be provided and shall serve both tie down and hoisting provisions, where practicable. These provisions shall be capable of withstanding a vertical acceleration of 3g. The hoisting ring assemblies shall conform to drawing 48B7796 or 49B6564 as applicable. The air conditioner shall have a minimum of 4 collapsible hoisting rings on each corner to permit hoisting of the enclosure for accessibility and a minimum of 4 to permit hoisting of the complete dry unit from the chassis. The hoisting ring assemblies shall be located around the periphery of the unit in a horizontal plane. The hoisting rings can be used as the tie down ring assemblies. The location of the hoisting ring assemblies shall be further governed by hoisting considerations. For hoisting, the rings shall be located with respect to the center of balance in order to comply with the following requirements.

a. The air conditioner shall not tilt at an angle greater than 15 degrees when hoisted from a single point with 4 equal length cables attached to the ring assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

b. The angle of the air conditioner with the horizontal shall not increase when the air conditioner is hoisted from a single point with cable lengths that cause the unit to tilt at an angle of 30 degrees with the horizontal.

* 3.5.5 Air transportability. The air conditioner shall be air transportable in accordance with MIL-A-8421 for C130 and C141 aircraft.

3.5.6 Detachable fittings. AN or MS type disconnect plugs and receptacles in the electrical system and readily detachable and attachable type fittings in mechanical components shall be utilized wherever practicable to permit rapid component removal and replacement.

* 3.5.7 Radio interference. The air conditioner shall meet the radio interference requirements of MIL-STD-461, Class IIIC Test Method RE05. Conducted radio interference in power cable shall not exceed limits of MIL-STD-462, Class 1c Test Method CE05.

* 3.5.8 Noise level. The noise level generated by the air conditioner shall not exceed the values as specified in MIL-STD-1474 Table 2, Category D, when measured as specified in 4.6.14.

* 3.5.8.1 Sound attenuation. As required, double wall type construction shall be used for the air intake plenum, air discharge plenum, the entire air conditioner enclosure and subenclosure. When double wall construction is utilized the acoustical material between the inner and outer wall shall be $1\frac{1}{2}$ lb density fiberglass insulation per MIL-I-22023 or equal. The perforated inner panel for air intake and air discharge plenum shall be in accordance with AISI 304L or 316L stainless steel. The perforated inner panel shall have 3/32 inch diameter holes on 3/16 inch staggered center. The perforated inner panels shall be mounted in such a way to not permit acoustic energy to be transmitted through fasteners to the outer panels. No external means shall be used to suppress the noise level.

3.5.9 Accessibility. The air conditioner shall be so designed and constructed that the following actions may be accomplished by removing a limited number of components and/or subassemblies (other than panels):

a. Remove the prime mover by removing only the mounting bolts, disconnecting fuel lines, control cables, V-belt(s) and lifting it out by the use of an overhead crane or forklift truck with the air condition enclosure removed.

b. Remove the compressor by uncoupling the flexible coupling, refrigerant plumbing, removing the mounting bolts and lifting it out the top, side or end of the air conditioner by the use of an overhead crane or forklift truck.

c. Remove the blower by removing the V-belt(s), the flexible duct connections and mounting bolts and lifting it out of the top side or end of the air conditioner by use of an overhead crane or forklift truck.

d. Remove the condenser motor by removing the mounting bolts, V-belts, disconnecting the electrical wiring and lifting it out the top, side or end of the air conditioner by the use of an overhead crane of forklift truck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

3.5.10 Safety features.

3.5.10.1 Provisions for the safety of personnel shall be incorporated to the maximum extent possible with regard to anticipated operating conditions and the capability of operating personnel. The danger to personnel from fire and explosion shall be avoided by separation of hazardous substances from heating sources, incorporation of vents and drains and other fire prevention measures.

3.5.10.2 Guards. Guards shall be provided on the air conditioner to shield rotating or hazardous components. Where required, the guard shall be removable to permit maintenance and servicing.

3.5.10.3 Belt drives. All belt drives used in the air conditioner shall utilize belts which are oil resistant and do not conduct static electricity.

3.6 Performance

3.6.1 Ambient operating range. The air conditioner shall be capable of operating within the range of ambient conditions described on a psychrometric chart by the following limits:

a. The 0° Fahrenheit (F) dry bulb line to intersect with its saturation line.

b. The saturation line from 0° to 90° F.

c. A line from "90° F" dew point on the saturation line to 140° F at 0 percent relative humidity.

3.6.1.1 Operating and storage capability. The air conditioner shall be capable of operating as a ventilating blower, without refrigeration, at ambient temperature between 0° F and +125° F. It shall be capable of being stored at -65° F to +165° F without detrimental effect or maintenance prior to being put into service.

3.6.1.2 Delivery pressure. At standard atmospheric pressure, without manual adjustment, the air conditioner shall deliver air at the specified external static pressure ranging from 0 to 3 pounds per square inch gage when measured at the end of a 15-foot section of duct.

* 3.6.1.3 Rated condition. When operating at a sea level ambient condition of 100° F ±3° F dry bulb (DB) and 85° F ±3° F wet bulb (WB), the air conditioner shall be capable of delivering 45 pounds per minute of air at temperature of 50° F ±3° F and a minimum of 90 percent relative humidity air against a maximum external static pressure of 3 pounds per square inch gage.

3.6.1.4 Design summer day condition. When operating in an ambient condition of 100° F ±3° F DB and 76° F ±3° F WB, the air conditioner shall be capable of delivering a minimum of 60 pounds per minute of 50° F ±3° F against a maximum external static pressure of 3 pounds per square inch gage. Under these conditions, the air conditioner shall also be capable of delivering 50° F ±3° F air within ten minutes after prime mover start.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.6.1.5 Design hot day condition. When operating in an ambient condition of 125° F ±3° F DB and 75° F ±3° F WB, the air conditioner shall be capable of delivering a minimum of 50 pounds per minute of 50° F ±3° F against a maximum external static pressure of 3 pounds per square inch gage.

3.6.1.6 Design humid day condition. When operating in an ambient condition of 97° F ±3° F DB and 87° F ±3° F WB, the air conditioner shall be capable of delivering a minimum of 45 pounds per minute of 50° F ±3° F air against a maximum external static pressure of 3 pounds per square inch gage.

3.6.1.7 Low ambient condition. When operating in an ambient condition of o° F, the air conditioner shall be capable of delivering a minimum of 45 pounds per minute of 50° F ±3° F against a maximum external static pressure of 3 pounds per square inch gage.

3.6.1.8 Designed high altitude hot day. When operating in an ambient condition of 5000 feet (12.23 pounds per square inch absolute maximum barometric pressure), 107° F ±3° F DB and 87° F ±3° F WB, the air conditioner shall be capable of delivering a minimum of 45 pounds per minute of 50° F ±3° F air against a maximum external static pressure of 3 pounds per square inch gage.

3.6.2 Titled position. The air conditioner shall be capable of operating satisfactorily when its normal operating plane is an angle of 8 1/2° in any direction from true horizontal.

3.6.3 Environmental exposure. The air conditioner shall be capable of conforming to all requirements specified herein after exposure to the following environmental conditions:

a. Ambient temperatures of -65° F to +165° F.
b. Radiant energy as found under natural conditions.
c. Salt spray as may be encountered in salt-sea atmosphere.
d. Fungus growth as may be encountered in tropical climates.
e. Rainfall as encountered in any locale.
f. Sand and dust as may be encountered in desert areas.
g. Altitudes from sea level (14.7 psia) to 5,000 ft (12.23 psia)

3.7 Refrigeration system. The air/refrigeration system shall include a refrigerant compressor, condensers, receiver, evaporator (aftercooler and precooler) controls and accessories.

3.7.1 Refrigerant. The refrigerant shall be dichlorodifluromethane (R-12).

* 3.7.1.1 Testing, evacuation and charging. Prior to charging with refrigerant, the system excluding the compressor shall be evacuated to a minimum of 0.5 mm Hg absolute. The system shall be held at this evacuation pressure for six hours and the evacuation pump shall be operating during this six hour period. If no noticeable pressure rise occurs, the unit may be charged with refrigerant. The evacuation shall be performed while the unit is in prevailing indoor ambient temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.7.2 Compressor assembly. The compressor shall be of the multi-cylinder reciprocating type conforming to good refrigeration practice and shall have sufficient capacity to meet the air conditioner requirements specified herein when operating at 1750 revolution per minute (RPM). The compressor shall be indirectly driven by V-belt(s) from the electric motor. The V-belt(s) shall be sized for the application. The compressor shall be procured from the compressor manufacture and shall be equipped with suction and discharge valves, that can be rotated 360 degree without interference, three electric solenoid operated capacity control actuator assemblies, internal cylinder unloading mechanism assemblies so the compressor can start without load and its capacity can automatically vary in steps according to the load on the evaporator. The compressor shall be capable to withstand a minimum of 350 psig on the low side and a minimum of 500 psig on the high side. Capacity control actuators shall be 24 volts/60 Hertz/1 phase. The compressor shall be provided with a readily observed oil level sight glass and shall contain a charge of refrigerant oil in accordance with VV-L-825. The compressor shall be the Trane Company reciprocating Series E or F.

3.7.2.1 The compressor and compressor motor assembly. The compressor and compressor motor assembly shall both be mounted on a common structural component of the unit of adequate rigidity to ensure alignment. The compressor and compressor motor shall be directly coupled in such fashion that either can be readily replaced without disturbing the other and without removal of the refrigerant plumbing.

3.8 Condenser coil assembly. The condenser shall be of the air cooled type conforming to good refrigeration practice and shall have adequate surface and air supply to enable the unit to meet the maximum capacity requirement specified in 3.6.1.4 with a refrigerant condensing temperature not exceeding 130° F (head pressure of approximately 180 pounds per square inch gage) and a refrigerant condensing temperature of 150° F (head pressure of approximately 235 pounds per square inch gage) for capacity requirements as specified in 3.6.1.5.

* 3.8.1 Condenser coil. The condenser coil shall be of the tube-finned type constructed of aluminum fins and copper tubes. The condenser coil shall have a minimum of 12 fins per inch and shall be integral with or securely bonded to the tubes by a continuous metallic bond. The coils shall be properly circuited, without traps and headed to provide the capacity required at minimum pressure drop. The aluminum fins shall have minimum 0.006 inch thickness with the surface corrugated and the edges straight. The coil shall be adequate size to limit the air velocity through the coil to 750 feet per minute.

3.8.1.1 Mechanical bonding. Mechanical bonding of the fins to the tubes will be acceptable provided that fins with integral collars are used and the tubes are expanded into the fin collars. Fin collars shall completely sheath the tube and shall be free from splits, cracks, or other defects. An interference fit of 0.005 to 0.008 inches shall be provided after final assembly of the fins and tube.

* 3.8.1.2 The condenser components. The condenser components shall be suitably cleaned prior to assembly and be free of oil or other foreign matter during assembly.

Source: https://assist.dla.mil --Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

3.8.1.3 The condenser coil protection. The condensing coil, if exposed to the exterior of the unit, shall be protected with a suitable grating or screen with an opening no larger than 3/4 inch. The protective device shall be spaced 2 inches from the condenser fins.

3.8.2 Condenser cooling fan(s). The air conditioner shall be equipped with a commercial type statically balanced condenser fan(s) so located as to draw ambient air through the condenser and discharge air through the top of the air conditioner in a vertical direction without recirculation. The fan(s) shall be directly driven or V-belt driven by electric motor(s). If the condenser fan is V-belt(s) drive, it shall be so designed to facilitate V-belt(s) removal/replacement and shall have adequate means for V-belt(s) adjustment. The condenser fan(s) shall deliver adequate airflow to maintain this condenser head pressure below 200 psig on a design hot day conditions specified in paragraph 3.6.1.5.

* 3.8.3 Condenser discharge sliding door. The air conditioner shall be equipped with a condenser discharge sliding door. The sliding door shall have handles on each side, right angle to the direction of the slide and shall cover the condenser discharge fan opening when the air conditioner is not in operation. The sliding door shall be weathertight to prevent rain and other foreign debris from entering the unit. The sliding door assembly shall permit personnel on the ground to operate the door smoothly when opening and closing. The condenser sliding door shall be equipped with a minimum of eight nylon rollers, four on each side of the condenser sliding door adequately sized for the application. The rollers shall be enclosed in channels to assure smooth operation of the condenser door. The door shall have a latch to hold it securely in the closed position.

* 3.9 Receiver. The air conditioner shall be provided with a refrigerant receiver of a size to hold the complete charge of refrigerant after pump down. The refrigerant charge shall not be greater than 80 percent of the receiver volume. The receiver shall be constructed in accordance with ASME Unfired Pressure Vessel Code requirements. The maximum working pressure and temperature of the receiver shall be 300 psi and 180° F respectively.

* 3.9.1 Receiver components. The receiver shall be equipped with adequate wrench type service valves on the inlet and outlet connections. For entrapping the total refrigerant charge in the receiver. The receiver shall be equiped with an automatic relief valve and shall not be capable of being capped. Flanges shall be provided as close as possible to the receiver. One flanged connection shall be installed upstream and adjacent to the inlet wrench type service valve and one flanged connection shall be installed downstream and adjacent to the outlet wrench type service valve. A drain plug shall be provided at the bottom of the receiver.

* 3.10 Evaporator coil assembly. The evaporator(s) (aftercooler and precooler) shall be of the finned type constructed of aluminum fins and copper tubes. The aftercooler coil shall have a minimum of 12 fins per inch and the precooler coil shall have a minimum of 8 fins per inch and shall be integral with or securely bonded to the tubes by a continuous metallic bond. The coils shall be properly circuited, without oil traps and headed to provide the capacity required at minimum pressure drop. The aluminum fins shall have a minimum 0.006 inch thickness with the surface corrugated and the edges

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

straight. The coil shall be of adequate size to limit the air velocity through the aftercooler coil to 500 feet per minute at maximum capacity requirements specified in 3.6.1.4.

3.10.1 Mechanical bonding. Mechanical bonding of fins of the tubes will be acceptable provided that fins with integral collars are used and the tubes are expanded into the fin collars. Fin collars shall completely sheath the tube and shall be free from splits, cracks or other defects. An interference fit of 0.005 to 0.008 inches shall be provided after final assembly of the fins and tubes.

* 3.10.2 Evaporator components. The evaporator component shall be suitably cleaned prior to assembly and free of oil or other foreign matter during assembly.

* 3.10.3 Condensation pan and drain. A condensation pan and drain shall be provided for each evaporator coil sections (precooler and aftercooler coil) and a suitable water eliminator shall be provided for the aftercooler evaporator coil sections to collect and dispose of moisture condensed by the evaporator coils. The drain shall be provided with anti-siphoning device to insure condensate drainage while the air conditioner is operating. The water eliminator and condensate pan shall be constructed of AISA 304 or 316 stainless steel. The condensate drains of each evaporation section (precooler and aftercooler) shall be manifold together with a proper slope to assure drainage. The manifold shall be fabricated from copper and shall terminate at a convenient location outside the air conditioner with a male hose connection. The male hose connection shall be adequately supported to the air conditioner. The male hose connection shall have a cap to seal off the opening when not in use. The cap shall be attached to the air conditioner by a retaining chain of adequate length.

* 3.10.3.1 Condensate drain hoses. Condensate drain hoses and all necessary fittings shall be provided to carry away condensate water from the air conditioner. The hose and fitting shall not be damaged by exposure to the temperature as specified in 3.6.3. The hose shall be 25 feet long and be of an adequate size to carry away the condensate water encountered during the operation for each condition as specified in 3.6.1.3 through 3.6.1.8. The 25 foot long hose shall be stored in a compartment within the enclosure of the air conditioner.

* 3.10.3.2 Condensate drain marking. A condensate drain connection marking shall be printed in a easily observed location near the drain connection.

3.11 Air/refrigeration control system and accessories. The air/refrigeration control system shall be equipped with liquid line solenoid valve(s), liquid charging valve, dryer strainer, sight glass, liquid sub-cooler, hot gas bypass system, suction line throttling valve, head pressure control system, refrigerant pressure safety controls, thermostats, and check valve(s).

3.11.1 Air/refrigeration control system. The air/refrigeration temperature control system shall consist of two subsystems to control the air temperature leaving the pre-cooler and after cooler evaporator coil. Controls of the precooler evaporator coil (C1) shall be accomplished by a thermostat

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

(T1) which senses entering ambient air temperature. The thermostat shall close on temperature increase at 73(±3)°F, energizing the pre-cooler liquid line solenoid valve (S1) and the compressor first state cylinder unloading valve (U1), thereby increasing refrigeration capacity by bringing the pre-cooler "on line." The thermostat shall open on decreasing temperature at a maximum of 6°F below the temperature increase setpoint. At this point S1 and U1 shall be de-energized and coil C1 shall be taken "off-line." Control of the after-cooler evaporator coils (C2, C3 and C4) shall be accomplished by a subsystem consisting of four major components: (a) temperature control valve (TCV); (b) temperature controller (TC); (c) air temperature sensor (TS); and potentiometer (R1). The TS shall be located in the air duct downstream from the after-cooler evaporator coils and shall sense the air temperature leaving coil C4. R1 shall be located on the operator panel and shall be used by the operator to select the desired delivered air temperature. The TCV shall modulate the refrigerant flow through the after-cooler coil C4, as directed by the TC, to match the evaporator capacity to the existing load. The TC shall receive input signals from the TS and R1 and shall modulate the TCV, energize or de-energize the compressor cylinder unloader valves (U2 and U3), energize or de-energize the liquid line solenoid valves (S2 and S3) to the aftercooler coils C2 and C3 and energize or de-energize the hot gas bypass and quench valve solenoids (S5 and S6) as necessary to maintain the desired delivered air temperature. The system shall control the delivered air temperature to the desired setpoint (50° to 80°F) within the allowable temperature band (±3°F).

Lists of nomenclatures for reference designators.

Reference Designator	Nomenclature/Description
C1	Precooler Coil
C2, C3	First two-thirds of Aftercooler Coil
C4	Last one-third of Aftercooler Coil
R1	Leaving Air Temperature Setpoint controller Potentiometer (50° to 80°F)
S1	C1 Solenoid Valve
S2	C2 Solenoid Valve
S3	C3 Solenoid Valve
S5	Hot Gas Bypass Solenoid Valve
S6	Quench Solenoid Valve
T1	C1 Entering Air Temperature Thermostat Switch
TC	Electronic Temperature Controller
TCV	Temperature Control Valve
TS	Leaving Air Temperature Sensor
U1	C1 Unloader Valve
U2	C2 Unloader Valve
U3	C3 Unloader Valve
V2	Quench Valve
V3	Hot Gas Bypass Valve
X1	C1 Thermostatic Expansion Valve
X2	C2 Thermostatic Expansion Valve

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

X3	C3 Thermostatic Expansion Valve
X4	C4 Thermostatic Expansion Valve

* 3.11.2 Solenoid valve. Solenoid valve(s) shall be provided ahead of each expansion valve(s) to allow pumpdown of the refrigeration system prior to the stopping of the compressor. A switch on the control panel shall de-energize the solenoid(s) valve(s). This will allow the compressor to operate until suction pressure actuates the low pressure cutoff switch stopping the compressor and condenser fan. All necessary pressure controls and relays to insure automatic pumpdown shall be provided. The automatic pumpdown shall not affect the operation of the unit in a ventilation mode. Soleniod valve shall be installed upstream of the refrigerant compressor, and a check valve shall be installed down stream of the refrigerant compressor to prevent refrigerant migration back to the compressor sump.

* 3.11.3 Charging valve. A service valve shall be provided upstream of the dryer-strainer for charging refrigerant into the system.

3.11.4 Dryer/strainer. A replaceable element type refrigerant dryer-strainer of sufficient capacity shall be provided upstream of the sight glass. The dryer shall be of sufficient size such that the moisture content of the refrigerant after charge into the unit is no greater than 15 parts per million by weight.

3.11.5 Sight glass. A line or bypass sight glass shall be provided in the liquid line located where the condition of the refrigerant may be observed while the air conditioner is operating and while charging the system with refrigerant. Protective caps shall be provided to cover the sight glass during transit and other periods of nonoperation.

3.11.6 Thermal expansion valve. The evaporator(s) (aftercooler and precooler) shall be equipped with an equalizer type thermostatic expansion valve(s) of sufficient capacity to meet the requirements specified herein.

* 3.11.7 Subcooler. A subcooler shall be provided in the liquid line of the refrigeration system. The subcooler shall be provided upstream from the expansion valve(s). This will prevent liquid refrigerant from flashing prior to entering the expansion valve(s).

3.11.8 Hot gas bypass system. A hot gas bypass system shall be incorporated in the air conditioner to provide an artificial load on the last two cylinders of the compressor upon sensing a decrease in evaporator load (aftercooler). The hot gas bypass system shall maintain continuous compressor operation down to and including 0° F. A liquid line with a liquid injection valve and a hot gas bypass line with solenoid valve and a hot gas bypass valve shall be provided. The two lines shall feed into compressor suction line to maintain an acceptable suction pressure through the range of capacity loads. The pressure drop shall not cause excessive superheat or the introduction of refrigerant into the suction line thus preventing damage to the compressor. A liquid injection valve, hot gas bypass valve and solenoid valve shall be provided and properly sized for the application. The solenoid valve shall be ahead of the hot gas bypass valve. The electrical characteristics of the solenoid coil shall be 24 volts/60 Hertz/1 phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.11.9 Head pressure control system. A head pressure control system shall be incorporated in the air conditioner to allow continuous compressor operation down to and including 0° F ambient temperature. The head pressure control system shall incorporate a 3-way head pressure control valve and a pressure control fan cycling switch to maintain adequate compressor discharge pressure for satisfactory operation down to and including 0° F ambient temperature. The pressure control fan cycling switch shall be located in the main control panel.

* 3.11.10 Pressure safety controls. The compressor shall be equipped with an oil safety switch, low pressure switch and high pressure switch. These safety controls shall be located in the main control panel.

3.11.10.1 Oil safety switch. The oil safety switch shall stop the compressor if there is a lubrication failure due to oil leaks from the system, a clogged strainer stopping the oil intake to the pump, excessive refrigerant in the crankcase or insufficient oil pressure. The oil pressure switch shall have a manual reset and a time delay switch to prevent inadvertent or nuisance stoppage due to momentary fluctuation of oil pressure.

3.11.10.2 Low pressure switch. The low pressure switch shall stop the compressor when the suction pressure is reduced to a point which could produce freezing of the coils or which would permit operation beyond the prescribed operating limits of the compressor. The low pressure switch shall be of the automatic reset type.

3.11.10.3 High pressure switch. The high pressure switch shall stop the compressor when the discharge pressure rises above the prescribed limits because of inadequate condensing, an overcharge of refrigerant, or air in the system. The high pressure switch shall be of the manual reset type.

3.11.10.4 Pressure gage. Pressure gages shall be of the flush mounted dial type having a minimum dial diameter of 2 1/2 inches. The gages shall conform to MIL-G-8402 as applicable and shall have an accuracy of at least $\pm$2 percent of full scale range.

* 3.11.10.5 Temperature controller. A temperature controller shall be provided to control the leaving air temperature. The leaving air temperature shall be manually selected and automatically maintained during the air condition operation. A small electric heating element shall be attached to the leaving air temperature sensing bulb located in the leaving air plenum which is controlled by a rheostat located on the control panel. The thermostats and suction line throttling valve shall vary the leaving air temperature from 50° F $\pm$3°F to 80°F $\pm$3°F when the ambient temperature is down to and including 0° F.

* 3.11.10.6 Suction line throttling valve. A suction line throttling valve shall be provided to control the leaving air temperature by throttling the suction pressure to balance the compressor capacity to the evaporator (aftercooler) load. The suction line throttling valve sensing bulb shall be located in the leaving air plenum.

3.11.10.7 Pump down control. Automatic pump down of the refrigerant system shall be controlled by the use of solenoid valve in the hot gas bypass

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

stream of the dryer and strainer. The solenoid valve shall be actuated through the compressor controls. Opening of the soleniod valve shall coincide with compressor start up. Valve shutoff shall permit continued operation of the compressor until the suction pressure is reduced to a value that will actuate the low pressure safety control switch stopping the compressor motor.

3.12 Refrigerant piping. All refrigerant piping shall be Type L copper in accordance with WW-T-799 except the vibration eliminators shall be as specified in paragraph 3.12.1. The piping shall be fabricated with the use of high temperature solder conforming to QQ-B-654. Flared end connections shall be used where disassembly for component replacement is required; however, such applications shall be in accordance with ASHRAE Guide and Data Book.

3.12.1 Vibration eliminators. The refrigerant compressor lines shall be equipped with vibration eliminators to control transmitted vibration and allow removal of the compressor without the breaking of the soldered joints.

3.12.1.1 Vibration loops. Vibration loops shall be incorporated in the lines to prevent the transmission of vibration to pressure control devices and gages. All refrigeration controls and piping shall be rigidly fastened to supporting structure.

* 3.12.1.2 Component removal. To facilitate removal of components from the power pac, all refrigeration lines, capillary lines and electrical wires etc., shall not be clamped to the power pac or the air conditioner enclosure.

3.13 Air flow system. The air flow system of the air conditioner shall include and inlet filter, evaporator, blower, evaporator(s) housing (aftercooler and precooler), high airflow protection switch, rigid duct work, discharge and intake openings and flexible air duct.

* 3.13.1 Filters. The air conditioning unit shall be equipped with suitable air filter(s) and shall be a National Bureau of Standard average efficiency of 72 percent installed on the ambient air intake to the unit. The filters shall be of the permanent, cleanable, reusable type not subject to corrosion by moisture and salt laden atmosphere and shall be readily removable for cleaning. Filters used in the conditioned air intake opening shall be 2 inch nominal thickness and the maximum face velocity shall be 500 feet per minute. Filters used in air intake openings supplying only ambient air to components, condensers, etc, shall be of the one inch nominal thickness with a maximum face velocity of 500 feet per minute. Where more than 1 filter is required, all filters on each type shall be of the same size. The aluminum frame and media shall be of adequate rigidity to prevent any deformation of the filter such as twisting, bulging, etc, during use and handling.

* 3.13.2 Blower. A centrifugal type blower of adequate capacity to meet the requirements of 3.6.1.2 through 3.6.1.8 shall be provided when operating at 3550 (revolutions per minute) input. The blower shall be dynamically balanced for the selected RPM of operation.

* 3.13.2.1 The blower and blower motor shall be mounted on a common structural component of the unit of adequate rigidity to assure alignment, and connected in such a fashion that either can be readily replaced without removal off rigid ducting. The blower shall draw ambient air through the air intake filter, evaporator coil (precooler) and deliver through the evaporator coil (aftercooler) and to the flexible duct.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.13.2.2 Blower flexible conections. The blower plenum outlet and inlet shall be connected to the air system by means of a flexible connection to prevent mechanical transmission of the blower vibration to the system.

3.13.2.3 Blower Lubricant fill/drainage . The blower shall incorporate all plumbing and valves to facilitate lubrication/fill and drainage. The drain tube shall be easily accessible from beneath the fill chassis.

3.13.4 Evaporator housing(s). The evaporator housing(s) for the aftercooler and precooler shall be installed in the duct system. The precooler housing shall be up stream of the blower to cause the air to be drawn through the precooler coil. The aftercooler shall be downstream of the blower to cause the air to be blown through the aftercooler coil. Provisions shall be made to allow for the removal of the evaporator housing(s) from the duct work.

* 3.13.5 Air damper assembly. An air damper assembly shall be incorporated in the air conditioner on the downstream of the evaporator coil (precooler) and upstream of the centrifugal blower. The air damper shall consist of a set of movable vane(s) controlled through mechanical linkages actuated by a damper motor. Positioning of the air volume control setting mounted on the control panel shall operate the damper motor to vary the air volume delivered.

* 3.13.5.1 Air damper motor. An air damper assembly motor shall be incorporated in the air conditioner of sufficient size to meet the requirements specified herein. The air damper motor shall be of the modulating type actuated by the air volume control which in turn shall open and close the damper vanes. The electrical characteristics shall be 24 volts AC/60 Hertz/1 phase.

3.13.5.2 Maximum Airflow Control. A means shall be incorporated into the design of the airflow system to limit the conditioned airflow to a value which will prevent blower prime mover overload.

* 3.13.6 Rigid ductwork. All internal ducts, duct transitions, plenums, etc, shall be provided in the enclosure for guiding air through the various assemblies. The ducts shall be constructed of material to withstand 5 1/2 psig, and bends and turns shall be internally smooth. Where applicable, duct intersections shall insure proper uniform mixing of various airstreams. All duct joints shall be accomplished with mating companion flanges properly gasketed. A double wall construction shall be a perforated sheet metal with 3/32 inch diameter holes on 3/16 inch staggered centers. The acoustical material contained between the inner and outer wall shall be as specified in paragraph 3.5.8.1.

3.13.7 Discharge and intake openings. The air conditioner shall have a single opening for the air entering and two discharge openings leaving the conditioned air circuit. The discharge opening shall be provided on the external wall of the cabinet and shall incorporate four male couplings in accordance with drawing 64C1004 with provisions for attachment of the ducts specified herein. Also, a cap in accordance with Air Force drawing 8044080 shall be provided to seal off one of the openings when not in use. The cap shall be retained to the air conditioner by a cable or chain of adequate length. All air intake openings in the air conditioner such as condensers,

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

and air supply shall be protected by means of heavy wire or expanded metal screen of 1/2 to 3/4 inch mesh to prevent entrance of large foreign objects. A distance of 2 inches shall be maintained between the screen and filter(s).

3.13.8 Flexible air ducts assemblies. The air conditioner shall be equipped with four 8-inch nominal diameter flexible ducts assemblies , in accordance with Air Force drawing 8321218. The air conditioner shall contain adequate space for storing the duct assemblies. On one end, the ducts shall incorporate a female connector conforming to Air Force drawing 64C1005 so that the ducts can be connected in series. The female end of the duct shall connect to the air conditioner discharge opening.

3.13.8.1 Flexible air duct color. The color of the exterior surface of the flexible air duct shall be HL (olive drab) in accordance with MIL-STD-808.

* 3.14 Prime mover assembly

* 3.14.1 Prime mover assembly. The prime mover for the air conditioner shall be driven by motor(s) designed for 230/460V/3 phase 50/60 HZ, 4 wire, operation (see 3.16.1) the air conditioner shall be wired for 280V/460V 50/60 HZ power when delivered.

3.15 Cabinet trailer design features.

3.15.1 General. The air conditioning unit shall be mounted on a four wheel trailer and shall be provided with parking brakes, suspension system, towbar and fenders. Requirements of MIL-M-8090 Type I, Group C, Class 2 shall apply except as modified herein.

* 3.15.1.1 Chassis. The trailer chassis shall be of welded steel construction and shall meet the structural requirements of paragraphs 3.5.3 and 3.5.4. The trailer shall not employ system components for structural carry through, but shall supply all structure integral in the chassis proper.

* 3.15.1.2 Power pac components. All components of the power pac such as the diesel engine, compressor, blower and alternator shall be mounted on a welded rectangular frame fabricated from steel of sufficient size to prevent deflection which could impose detrimental stresses on the components. The power pac assembly including the rectangular frame shall be mounted on the steel base of the air conditioner with adequate resilient mounts to minimize vibration.

* 3.15.1.3 Clearance. The fully loaded trailer shall have a road clearance of at least 7 1/2 inches at all points with tire pressure of 100 pounds per square inch. The trailer shall have sufficient tire clearance for use of tire size 7.50-10.

* 3.15.2 Tie down ring assemblies. The air conditioning unit shall be equipped with tie down fittings to satisfy the requirements of paragraphs 3.5.4, 3.5.5, and MIL-A-8421. The tie down ring assemblies mountings and carry through structures shall also be designed to comply with these requirements. The air conditioning unit tie down arrangements shall be marked in accordance with MIL-S-8421. A total of eight tie down fittings conforming to Air Force Drawing 49B6564 or 48B7796 shall be provided. The tie down

Source: https://assist.dla.mil --Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

fittings shall be located near the corners of the unit, two on each side, and shall be secured to the main structural support frame. The fittings shall be so mounted that when the rings are raised to a vertical position for hoisting purposes, a minimum clearance of 1 1/4 inches between the side of the unit enclosure or cover and the inside of the hoisting ring is provided.

3.15.3 Running gear. The unit shall be provided with four wheel running gear in accordance with MIL-M-8090, Type 1, Group C mobility, except that maximum towing speed shall be 20 miles per hour (mph).

3.15.3.1 Wheels, hubs, and tires. The four wheels, hubs and ties shall be in accordance with MIL-W-8005. The rear wheel shall be equipped with parking brakes to hold the air conditioner in a fixed position during all operations and positions specified herein. The brakes shall be readily applied or released by means of a single manual control located at the front of the unit in the vicinity of the draw bar. When set, the parking brakes shall be of sufficient capacity to hold the unit in a fixed position during a pull of up through 600 pound line pull, forward or backward, on dry level concrete. The vibration and shock encountered with usage under type mobility specified shall not cause the brakes to lock. The rear driving hubs shall be cast steel in lieu of cast iron. The tire size shall be 7.50-10.

* 3.15.3.2 Tire pressure marking. The tire pressure marking shall be as shown in figure 1 and shall comply with 3.18.4.

* 3.15.3.3 Steering. The front axle shall be equipped with a knuckle type steering incorporating the rods and kingpins to provide a high degree of maneuverability. Adequate means for lubrication shall be provided. Lubrication fittings shall conform to MIL-F-3541. The cramping angle shall be no less than 30° and shall comply with 3.15.8. The steering mechanism shall be of the compensated type. The front axle and steering mechanism shall be able to withstand the loadings encountered during usage under the type mobility specified in 3.15.3.

3.15.4 Suspension system. The suspension system for the air conditioning unit shall be designed to meet all mobility, structural and proof load test requirements specified herein. The suspension system shall provide adequate springing to minimize acceleration inputs and provide a stable ride for the air conditioning unit. Means shall be provided to prevent occurrence of suspension frequencies falling into sympathy with the surface frequency. The suspension system shall not allow shock loads greater that 2 g's to be transmitted to the air conditioning unit under the mobility test conditions specified herein. The unit shall be so designed that it will not demonstrate an undesirable tendency to sway, skid, yaw, tilt or attain a resonant frequency.

* 3.15.5 Towbar assembly. The trailer shall be furnished with a hinged towbar and with a lunette eye conforming to MS51336 with a 3 inch inside diameter and a 1 5/8 inch diameter cross section. It shall be suitable for towing the air conditioner at the speeds specified herein or abruptly starting and stopping the unit without damage. The towbar, when stored in its folded position, shall not exceed above the maximum height of the unit. Suitable means shall be provided to retain the towbar assembly in the stored position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.15.6 Pintle. The rear end of the trailer shall be equipped with a pintle conforming to MS51335-2. The pintle shall be located 9 to 16 inches above ground level. The pintle hook stencil wording shall be in accordance with Figure 1.

3.15.7 Air condition enclosure. The entire air conditioning unit shall be encased in a weathertight enclosure that shall completely enclose all the components of the air conditioner. The enclosure shall be equipped with adequate access doors and panels for servicing and maintenance purposes. The air conditioner shall be designed to operate satisfactorily with all access doors and panels closed except for those access doors required to remain open during operation of the air conditioner such as condenser air intake doors, condenser discharge door, ambient air intake door/condition air discharge door, diesel engine radiator door and control panel door.

3.15.7.1 Enclosure frame. The enclosure frame shall be fabricated of structural steel members and shall be firmly attached to the trailer chassis to form an integral structure. Framing members may have removable sections as required for the removal of components.

* 3.15.7.2 Enclosure construction. The entire enclosure shall be constructed of Grade C carbon steel in accordance with QQ-S-698 with a minimum of 14 gage steel. All housing panels shall be secured to the framing members in such a way that their complete removal may be accomplished with a minimum of effort to facilitate servicing and maintenance. Panels and access doors frequently used shall be hinged and secured with rugged quick opening door latches of the tongue type utilizing a recess door handle flush with the cabinet. Catches for the access door unit shall be rounded off. Plastic knobs or handles shall not be used. When the access panel doors are open, they shall be secured so that high winds will not inadvertently close them from the open position or cause structural damage. As a minimum, hinged access doors shall be provided for access to the main junction box and the refrigeration service accessories. All gasketing used to provide weathertight seals for access doors, panels, etc, shall be suitable for the use intended. Access panel doors hinged at the top shall be avoided. Drain holes with a minimum of 0.25 inch in diameter or drain channels with run off tubes which would allow the escape of standing or entrapped water from the interior areas of the air conditioner shall be provided. Such areas include the shelves under the condenser racks where the flexible ducts are stored, base of the air conditioner where enclosures were formed to support components, etc. All metal joints formed in the fabrication of the air conditioner shall be so designed to prevent moisture and water entrapment.

* 3.15.7.3 Center of balance. The center of balance of this unit shall be identified by stenciling a "center of balance" marking on each side of the unit (see Figure 1). The center of balance of the air conditioner shall be with all components installed and the unit fully charged with R-12 refrigerant. The center of balance shall be determined by having the air conditioner tilted by 10° fore and aft during weighing operations and intersection of the respective vertical lines obtained.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.15.8 Maneuverability. The trailer shall be capable of executing a full turn to the right and to the left while being towed at a speed of 10 mph by a commercial type vehicle and shall not capsize nor skid during turns on dry concrete or macadam at this speed. The trailer shall be capable of turning within a radius of 17 1/2 feet. The turning radius is measured from the center of the circle to the outer edge of the inside wheel.

3.15.9 Mobility. The trailer shall withstand towing over a concrete or macadamized road at speed of not less than 20 mph and over loose sand or gravel road or upgrated sodden field at a speed of not less than 10 mph. Also, the trailer shall be capable of ascending, negotiating and descending a ramp having a slope of 20°.

* 3.16 Electrical system

* 3.16.1 General. The air conditioning unit shall be designed for 230/460V, 3 phase, 50/60 cycle, 4 wire operation unless otherwise required due to special power requirements. The instantaneous starting current for each motor shall not exceed 316 amps for 460V operation and 410 amps for 230V operation; the peek current for the unit during starting shall not exceed 400 amps for 460V operation or 570 amps for 230V operation. These amperages shall be exclusive of the first 0.05 seconds following actuation of each contractor. Starting and operating control circuitry shall be single phase 24 vac. The electrical system shall be fed by a power cable terminating at a main service breaker switch located on the unit. Operation shall be push button start and stop switches. A switch shall be provided for selection on "ventilation" mode or "cooling" mode. Circuitry shall be such that motorized and other devices required for the cooling mode are not energized until the blower circuit is activated. Refrigeration system safety devices shall de-energize all power to the motors and refrigeration system components under abnormal operation conditions. A momentary contact service switch shall be provided to bypass the safety devices at the startup of the refrigeration system. Means shall be provided to permit pump down of the refrigeration system if desired. Indicator lights shall be provided to indicate external power "on", blower "on", and cooling system "on ". Change of components specified hereunder will be allowed if necessary due to special power requirements.

3.16.2 Electrical system. The air conditioner electrical circuits shall contain insulated electrical wiring capable of withstanding the operating and environmental conditions specified herein. All electrical wiring shall use copper conductors and shall have visible numerical coding for ready identification by maintenance personnel. Electrical wiring used in the air conditioner shall not exceed the recommended capacity rate stated in the National Electrical Code.

3.16.3 Service cable and main breakers. The unit shall be equipped with a 600 volt, 50 foot, six conductor rubber covered service cable conforming to MIL-C-5756 and terminated in a suitable main breaker switch on the side of the unit. The cable shall be securely and permanently attached to the air conditioner in a manner that will relieve stress at the terminals. The other end of the cable shall be provided with a molded rubber, polarized, watertight type plug conforming to MS27265-6. The three power conductors shall be selected for 230V operation. The conductor used for grounding purposes shall be not smaller than size AWG No 4 and the 48V AC dead cable leads shall be not smaller than size AWG No 12. A clearly visible red

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

indicator light shall be located adjacent to the breaker switch to warn personnel of the voltage involved and at a minimum shall contain "WARNING 230/460V". Cable storage and main breaker shall be external but within the profile of the unit and shall provide adequate protection to allow closing the cable access door when the power cable is fully extended.

3.16.4 Starting control circuit. The starting control circuit shall be in accordance with Figure 2. The starting relay(s) shall be operated by push buttons located on the control panel. The starting relay(s) and protective devices shall be operated on 24V supplied from a transformer that can be connected to operate on either 230V or 460V. The switch leg leads C_1 and C_2 of Figure 2 shall provide switching control to energize the relays at the power receptacle thus providing dead cable control. This circuit shall be used for no other purpose.

3.16.5 Starter and overload protection. A reduced voltage starter or other means of limiting the starting current shall be provided. The starters shall include adequate under voltage and over voltage protection.

3.16.6 Voltage conversion. Voltage conversion from 230 to 460 and vice versa shall be accomplished by changing control transformer taps, starter heater elements, and motor winding connections. When delivered, the air conditioner shall be connected for 460 operation. A reversible warning plate shall be permanently attached to the cable storage compartment door indicating the appropriate voltage (230 or 460V) at which the unit is to be operated as well as the applicable wiring connection.

3.16.7 Motors. The compressor motor, blower motor and condenser fan motors shall be incorporated in the air conditioner and shall be of the squirrel cage inductor type. The electrical characteristics for the compressor and blower motors shall be 40 and 40 HP, respectively, 230/460V, 3 phase, 50/60 cycle totally enclosed type, equipped with the ball bearings and shall conform to CC-M-1807, Type II, Class 3, Style E, Enclosure 2, Design K. The condenser fan motor shall be 7 1/2 HP, 230/460 volts, 3 phase, 50/60 cycle, totally enclosed type equipment with ball bearings and shall conform to CC-M-1807, Type II, Class 3, Style E, Enclosure 2, Design K. All motors shall be selected so that when operating under the maximum performance requirements specified herein, the motor shall not exceed rated capacity.

3.16.8 Motor starting switches. Each of the electric motors in the unit shall be equipped with a magnetic starting switch of the proper size and type. The magnetic switch shall provide under voltage and overcurrent protection and shall be of the "manual reset" type suitable for the use intended. All magnetic starters shall be equipped with 24 vac actuating coils to conform to the requirements of paragraph 3.16.10.

3.16.9 Control voltage transformer. The voltage employed for the magnetic controls of the motors specified in 3.16.8 and the control specified in 3.16.1 shall be 24V, 50/60 cycle, single phase. A stepdown transformer of capacity as required shall provide the control voltage.

Source: https://assist.dla.mil --Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.16.10 Junction box. The unit shall be equipped with a main junction box which shall include the motor magnetic switch. The junction box shall be completely weatherproof and provided with a captive cover. Means shall be provided to allow resetting of the magnetic switches without opening of the cover. The connecting cables to the junction box shall be provided with electrical connector conforming to MIL-C-5015. Connector shall be potted in accordance with MIL-W-22759.

3.16.11 Wiring. All wiring shall be accomplished in accordance with MIL-W-22759 and wires shall be equipped with end terminals conforming to Class I, MIL-T-7928. Wiring within the unit shall be number coded in accordance with MIL-W-22759 and shall be in strict accordance with all applicable wiring diagrams and operating, service or maintenance instruction manuals. All electrical connectors shall be marked in accordance with MIL-STD-195. All wires shall be in accordance with MIL-W-22759 and shall be selected for continuous duty current capacity in accordance with MIL-W-22759.

3.16.12 Circuit breaker. The condenser fan motor shall be protected from overcurrent by means of a circuit breaker. The circuit breaker shall be of the type which is nonsensitive to temperature and shall be capable of tripping out the condenser fan motor during overloading conditions and shall be equipped with a manual reset. The condenser fan motor circuit breaker shall be identified by stenciling 1 1/2 inch letters.

3.16.12.1 Electrical relays. Electrical relays shall be of the sealed type in accordance with MIL-R-5757.

3.16.13 Control panel. The air conditioner shall be provided with a recessed composite instrument and control panel on which all instruments, gages, switches, warning lights, etc, are mounted. The instruments and control panel and its accessories shall be readily accessible to operating personnel and all items on the panel shall be permanently and legibly marked. The panel shall be covered by a suitable weathertight cover. The control panel and instruments shall be shock mounted. Suitable light assembly(s) shall be provided for adequate illumination of the panel. Light(s) shall be controlled by an on/off control.

3.17 Instruments. The following major controls shall be mounted on the control panel to facilitate ease of operation by personnel. Other switches and operating mechanisms shall be installed where applicable to perform the various functions required in the unit operation.

a. Suction pressure gage - Indicates the pressure of the refrigerant gas as it enters the compressor.

b. Discharge pressure gage - Indicates the pressure of the refrigerant gas as it leaves the compressor discharge valve.

c. Compressor oil pressure gage - Indicates oil pressure in the compressor.

d. Cool switch - Energizes and de-energizes the liquid solenoid valve. When the valve opens, suction pressure will cause the low pressure cutout switch to close and energize the compressor motor starting the compressor operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

e. Airflow gage - Indicates volume of air leaving the unit.

f. Air volume control - Controls the amount of air leaving the unit.

g. Air temperature control - Allows the operator to set and maintain the temperature of the air as it leaves the unit.

h. Discharge air pressure gage - Indicates pressure of discharge air as it leaves the unit and enters the flexible duct.

i. Condition air temperature gage - Indicates the temperature of the condition air as it leaves the unit.

3.18 Identification.

* 3.18.1 Instruction plates. An instruction plate(s) made of noncorrodible metal, mimimum thickness of 0.025 inch, shall be attached by rivets suitably located on the internal surface of panel cover specified in 3.17. The wiring diagram, refrigeration cycle diagram and instruction for start up, operation and shutdown procedures shall be photo etched on the instruction plate. All components shall be identified with their reference designation for ease in troubleshooting.

3.18.2 Identification of product. All equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.18.3 Control and gage nameplates. All control switches, gages and pilot lights shall be properly identified with permanently attached nameplates of noncorrodible material, minimum thickness 0.025 inch with nomenclature photo etched on the nameplate.

* 3.18.4 Markings. The markings for the air conditioner shall be in accordance with Figure 1. All markings shall be black in color except where otherwise noted. The color shall be in accordance with FED-STD-595 color 27038 semigloss or 37038 luster gloss.

3.19 Accessory equipment

* 3.19.1 Handbook compartment. A handbook compartment shall be provided and shall be minimum of 1 x 9 x 11 1/2 inches. The compartment shall be located near the control panel and shall be provided with drain holes.

3.20 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part number.

3.21 Flexible coupling. Flexible coupling(s) mounted on the motor(s), blower and compressor shall be secured to the shaft using taper lock key bushing(s). With respect to GFE compressor the driven shaft is tapered and the taper lock bushing will not be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 3.22 Dimensions and weight. The overall size of the trailer mounted diesel driven air conditioner shall not exceed a width of 77 inches, a length of 120 inches (exclusive of the pintle hook as well as towbar), and height of 78 inches. The maximum dry weight shall not exceed 6000 pounds.

3.22.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the air conditioner, they shall be held or limited accordingly.

* 3.23 Belt tension idler assemblies. An inside belt idler assemblies shall be incorporated between the engine/blower and between engine/compressor on the slack side of the belt drive. The idler assemblies shall be of the type that turning the adjustment screws would permit the idler to move upward or downward, to apply necessary tension and for belt removal or replacement. Belt removal/replacement shall be accomplished without removing major components.

* 3.24 Belts. The belts selected for the belt drive assemblies shall be adequate size for the application. The belts selected shall be gauged typed or joined.

3.25 Pulleys. Pulleys mounted on the power take off, blower, alternator and compressor clutch shall be secured to the shaft using taper lock key bushing. Keys shall be secured by set screw(s) to prevent key migration. Other pulley(s) shall use standard key with set screws.

3.26 Finishes and protective coatings

* 3.26.1 Cleaning, painting, plating, chemical films and chemical treatment. All cleaning, painting, plating, chemical films and chemical treatment shall be in accordance with the general requirements and procedures as defined in MIL-STD-808 and will be accomplished in conjunction with the revised Table I of Air Force Drawing 7545352. Cleaning and conditioning of HRPO steel may be accomplished by phosphoric acid etch in lieu of sand blasting.

a.	Ferrous metals	SF100
b.	Cadmium or zinc plated surfaces	SF221
c.	Aluminum and aluminum alloys	SF300
d.	Magnesium alloys	SF400

* 3.26.3 Finishing colors. The painting shall be in accordance with the general requirements and procedures as defined in MIL-STD-808 and shall be accomplished in conjunction with the Table I of Air Force Drawing 7545352. The exterior and interior film coats shall be dark green (FED-STD-595, No. 24052). All insides and exterior surfaces of the air conditioner shall be painted except the evaporator coils (aftercooler and precooler), condenser coils, refrigeration piping, controls, accessories, lubricating fittings and tie down rings.

3.27 Workmanship

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

3.27.1 General. The air conditioner, including all parts and accessories shall be fabricated and finished in a workman like manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, radii of fillets and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts; and tightness of assembly screws and bolts.

3.27.2 Fabrication. Machining, drilling and forming shall be accomplished with the use of accurate templates, jigs or gages. Boxes, cases, shields, and compartment walls shall be fabricated by casting, drawing or bending and welding except that riveting or bolting may be used where ease of servicing of the air conditioner requires that a removable panel construction be used where the applicable stresses dictate the use of a strong aluminum alloy that does not provide a good weld.

3.27.3 Riveting. Riveting operations shall be carefully performed to insure tight and satisfactory heading of rivets.

3.27.4 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from pronounced backlash.

3.27.5 Cleaning. The air conditioner shall be thoroughly cleaned. Loose, spattered or excess solder or brazing flux, metal chips and other foreign material shall be removed during and after final assembly. Necessary actions will be taken to neutralize flux to prevent corrosion on solder or brazed joints.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure suppliers and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection and testing of the air conditioner shall be classified as follows:

a. First article testing (See 4.4)

b. Quality conformance inspection (See 4.5)

4.3 Test conditions.

4.3.1 Method of testing and rating. The method of testing and rating the air conditioner shall be in accordance with the ASHRAE Standard 37-69.

4.3.2 Apparatus.

Source: https://assist.dla.mil --Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.3.2.1 Positive counter. Wherever practicable, rotational speed shall be determined by means of directly coupled positive counter that will count the revolutions for not less than one minute. A combination unit that counts the revolutions while measuring the time required in connection with other data is preferred. Strobotachometers may be used provided data can be furnished to attest to their accuracy. Hand tachometers shall not be used.

* 4.3.2.2 Nozzles or orifices. Airflow shall be measured by means of smooth approach nozzles or flatplate orifices conforming to the latest edition of ASHRAE Standard 37-69. Other apparatus such as pilot tubes, rotameter, etc, may be used only when specifically approved by the procuring activity. The rate of airflow shall be expressed in terms of pounds per hour.

* 4.3.2.3 Psychrometric chart. Humidity shall be determined by means of an ASHRAE psychrometric chart (ASHRAE Guide and Data book) from readings of wet and dry bulb temperatures. Air velocity over the web bulb thermometer shall be in accordance with the ASHRAE Standard 37-69. The thermometers shall be so arranged that the air passing over the wet bulb will not subsequently pass over the dry bulb. The wick of the wet bulb shall be wet with distilled water only and shall be replaced whenever a perceptible quantity of oil, dirt incrustation of other foreign substances has accumulated thereon.

4.3.2.4 Mercurial barometer. Barometric pressure shall be measured by means of a mercurial barometer at the same elevation as the test. The pressure shall be corrected for the temperature of the scale and mercury and for variation in gravity resulting from the location of the barometer with regard to elevation and latitude. Readings obtain from aneroid barometers will not be accepted.

4.3.2.5 Thermocouples and potentiometers. Temperatures shall be measured by means of appropriately located thermocouples and properly calibrated potentiometers. Insofar as practicable, thermocouples shall be insulated from contact with other metals. Temperature shall be expressed in degrees Fahrenheit. Laboratory type mercury thermometers will be permissible if used correctly in the application.

4.3.2.6 Liquid manometers. Insofar as practicable, pressure and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon tube gages. Gages shall be of laboratory type having a dial diameter of not less than four inches.

4.3.3 Readings. Readings of all refrigerant pressures, condenser, evaporator and discharge air temperatures shall be recorded at intervals not exceeding 30 minutes during all tests specified herein.

4.3.4 Air conditioning unit capacity. The condition air temperatures, airflow rates, relative humidities and static pressure specified herein shall be measured at the end of one 15 foot length of duct specified herein. (Capacity rating points are specified for standard sea level atmospheric pressure.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.3.5 Preliminary run-in. Any preliminary run-in specified by the manufacturer shall be performed prior to submission of the air conditioner for testing. A certified log of all preliminary running time shall be included in the air conditioner log.

4.4 First article testing.

4.4.1 Test samples. Two complete air conditioners, meeting all requirements of this specification, shall be subjected to the first article tests specified in 4.4.3 as a means to expedite testing.

4.4.2 Test Report. Upon completion of the first article tests, a test report in accordance with MIL-STD-831 shall be prepared and two complete copies and one reproducible copy of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance and irregular functioning shall be identified by accumulated operating time, cycles, miles or position in the test procedure, as appropriate. Test conditions at the time of events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control or other procedures of leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information by this paragraph, but clear reference to the location of the data shall be included.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.4.3 First article tests. Two complete air conditioners, meeting all requirements of this specification, shall be allocated to the first article tests as follows and tested as described under 4.6.

Sample number 1	Sample number 2
Examination of product	Examination of product
Operational test	Operational test
Environmental tests	Air transportability
Duct test	Mobility test
Radio interference test	Tilted position test
Noise level test	Performance, low ambient endurance and high altitude tests

4.5 Acceptance test. The acceptance test shall consist of the following:

a. Individual tests (See 4.6.2, 4.6.3 and 4.6.4)

b. Sampling Plan and Tests (ee 4.5.2)

4.5.1 Individual tests. Each air conditioner shall be subjected to the following tests as described under 4.6:

a. Examination of product See 4.6.1.

b. Operational test See 4.6.2.

4.5.2 Sampling plan and tests

4.5.2.1 Periodic sampling. One air conditioner shall be selected at random from the first ten and one for each subsequent ten or fraction thereof produced and subjected to the following tests.

a. Acceptance performance tests (See 4.6.3)

b. Final Inspection (See 4.6.4)

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extend and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure indicates that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test Method

4.6.1 Examination of product. Two air conditioners shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, markings and finishes prior to start of any first article testing.

4.6.2 Operational tests. The air conditioner shall be subjected to and shall satisfactorily complete the following tests. Individual records of all operational tests shall be maintained.

4.6.2.1 The air conditioning unit shall be started and operated in the cooling cycle at the prevailing normal room or ambient temperature for not less than 2 hours. The measured or indicated cooling system airflow shall be greater than 50 pounds per minute (ppm) and the condition air temperature shall not exceed 50° F $\pm 3^{o}$ F.

4.6.2.2 The temperature, airflow and safety controls shall be checked to insure proper operation and stability.

4.6.2.3 A functional test on all operating components of the air conditioner including brakes, steering, switches, lights, doors, ducts and connectors shall be conducted to insure proper operation.

* 4.6.3 Acceptance performance test. Under ambient condition specified in 3.6.1.4, it shall be demonstrated that within 10 minutes after prime mover start, the air conditioner is capable of delivering 50 pounds per minute of 50° F $\pm 3^{o}$ F air at 3 psig. The air conditioner shall also be operated for 2 hours under these ambient conditions and shall meet all the requirements of 3.6.1.4. At the end of this test, all safety controls shall be checked for effectiveness. The air conditioner shall exhibit no malfunctions, leaks, rough operation, vibration or other irregularities as a result of this test.

4.6.4 Final inspection. After completion of the acceptance performance test, the air conditioner shall be visually inspected to the maximum practicable extent. Indication of physical breakdown or defects shall be cause for rejection.

4.6.5 Environmental tests.

4.6.5.1 The following environmental test shall be conducted in accordance with the specified procedures of MIL-STD-810 except as noted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

* 4.6.5.1.1 High temperature. The high temperature test shall be in accordance with Procedure I except that the air conditioner shall not be operated during the test. Upon completion of the test, the air conditioner shall be started and operated as specified in 4.6.6. The air conditioner shall be inspected for detrimental effects on finishes, seals, etc.

* 4.6.5.1.2 Salt spray. The salt spray test shall be in accordance with the specified procedure for a period of 50 hours except that the visual examination shall be conducted prior to electrical starting. The air conditioner shall be started and operated as specified in 4.6.6. A comprehensive record of the effects shall be maintained for future reference in determining whether the effects of the exposure to the salt spray were contributory to possible failures encountered during other tests.

4.6.5.1.3 Fungus treatment. Certifications from manufacturer(s) that nutrient containing components are fungus treated per MIL-V-173 instead of testing as outlined is acceptable.

* 4.6.5.1.4 Rain. The rain test shall be conducted in accordance with the procedure specified except that the air conditioner shall be operated for a period of 30 minutes while still under the conditions specified. The air conditioner shall continue to deliver cooling air during and after exposure to the test.

* 4.6.5.1.1 Sand and dust. The sand and dust test shall be conducted in accordance with the procedure specified. Upon completion of the test, the air conditioner shall be examined and operated as specified in 4.6.6.

* 4.6.5.2 Low temperature exposure. The dry air conditioner shall be placed in a test chamber and cooled to and maintained at a temperature of -65° F for a period of 72 hours. At the end of this time, the air conditioner shall be allowed to reach prevailing ambient temperatures and then serviced, examined and tested as specified in 4.6.6.

4.6.5.3 Capacity check. Upon completion of the environmental testing of Sample Number 1, this sample shall be subjected to a capacity check under the conditions specified in 3.6.1.4 and shall satisfactorily meet the requirements specified herein.

4.6.6 First article operational test. The air conditioner shall be started and operated for a period of 30 minutes and checked for correct operation of all components and controls. During this time, the air conditioner shall exhibit no malfunctioning, leaks, rough operation, excessive vibration or other irregularities.

4.6.7 Air transportability. The ability of the hoisting provisions and the air conditioner to withstand the equipment restraint criteria specified in 3.5.3 and 3.5.4 hoisting components analysis shall be for the main chassis, the internal roof structure and tie down rings. Equipment restraint criteria analysis shall be performed sufficiently to show that the structure integrity will be retained and that no permanent deformation will result from specified loading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.6.7.1 The equipment restraint criteria identified in MIL-A-8421 will be employed in the analysis of the following critical points of stress:

a. The mounting of compressor and compressor motor blower, and blower motor to the power pac frame.

b. The mounting of the power pac frame, control panel, evaporator coils, condenser fan motor, receiver and running gear to the main chassis or supporting structure.

Calculation will be sufficient to show that these major items will not become projectile when subjected to the specified loadings.

4.6.8 Flexible Ducts. The flexible duct shall be tested in accordance with MIL-D-38386 or certified by the manufacturer that the ducts meet the test requirements of the specification.

4.6.9 Mobility test. The first article trailer mounted air conditioner shall be towed over concrete or macadamized road at a speed not less than 20 miles per hour for a distance of not less that 100 miles and over a loose sand or gravel road or an ungraded sodden field at a speed of not less than 10 miles per hour for a distance not less than 70 miles. The air conditioner shall satisfactorily enter, negotiate and leave a ramp with a 20 degree slope without bottoming. Upon conclusion of the test, the air conditioner shall be examined and tested as specified in 4.6.3.

4.6.9.1 Turns. The first article mounted air conditioner shall be towed over dry concrete or macadam by a commercial type vehicle through 10 complete turns each to the right and the left at a speed of 10 miles per hour. The trailer shall turn within 17 1/2 foot radius without capsizing or skidding. The turning radius is measured from the center of the circle to the outer edge of the inside wheel.

4.6.9.2 Brake performance. The first article trailer mounted air conditioner shall be parked and coupled to a towing vehicle, the handbrake shall be applied. The air conditioner shall then be towed for a distance of 10 feet. A towing force of 600 pounds will be required to be applied before the wheels roll.

* 4.6.9.3 Referigerant system leak. The first article trailer mounted air conditioner refrigeration system shall be checked for refrigerant leaks by charging the refrigerant system with R-12 to 75 psig, then pressurize the receiver to 300 psig with dry nitrogen containing refrigerant gas detectable with a halogensensitive electronic leak detector. Test pressure tolerance shall be plus 5 psig and minus 0 psig. Calibrate the leak detector for 25 percent concentration of R-12 mixture and set this indicator to indicate a leakage rate of 0.5 ounce per year. Check the entire refrigerant system for leaks. Either of the following shall constitute failure of the test:

a. Deformation of any component in the refrigerant system.

b. Any detectable refrigerant leaks.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.6.10 Tilted position. The air conditioner shall be tilted in 4 different positions, 90 degrees apart, an angle of 8 1/2 degrees to the true horizontal plane. The air conditioner shall be tested as specified in 4.6.11 at the various positions.

4.6.11 Performance and endurance tests. The air conditioner shall satisfactorily pass a performance and an endurance test of 600 hours total duration with all access doors and panels closed except for those doors required to remain open during operation of the air conditioner such as condenser air intake doors, condenser discharge door, ambient air intake door, control panel door and diesel engine door. Once started, these tests shall not be interrupted by using the unit for any other purpose. In conducting these tests, operation of the air conditioner shall be continuous except for shutdown of the unit as necessary for servicing, changes in test conditions, minor repairs, and replacement of minor parts and for nonoperating period associated with shift scheduling of test personnel. The total test time shall be divided as follows:

a. A 100 hour performance test on cooling (20 cycles)

b. A 50 hour ambient test at 0° F (10 cycles)

c. A 400 hour endurance test on the cooling components (80 cycles)

d. A 50 hour high altitude test (10 cycles)

4.6.11.1 Performance of 100 hours. The 100 hours shall be conducted in cycles of five hours duration. Each cycle shall be as follows: Four hours of continuous operation, 15 minutes of safety control recommended by the contractor, 30 minutes shutdown of the unit and 15 minutes for the unit to stabilize prior to another cycle. At every cycle the air conditioner shall be tilted at a different position, 90 degrees apart, at an angle of 8 1/2 degrees to the true horizontal plane for satisfactory operation of all components during tilt.

4.6.11.2 Zero degrees fahrenheit low ambient test of 50 hours. The 50 hours shall be conducted in five hours duration. Each cycle shall be as follows: Four hours of continuous operation, 15 minutes of safety control recommended by the contractor, 30 minutes shutdown of the unit and 15 minutes for the unit to stabilize prior to another cycle. At every cycle the air conditioner shall be tilted at a different position, 90 degrees apart, at an angle of 8 1/2 degrees to the true horizontal plane for satisfactory operation of all components during tilt.

4.6.11.3 Endurance test of 400 hours. The 400 hours shall be conducted in cycles of five hours duration. Each cycle shall be as follows: Four hours of continuous operation, 15 minutes of safety controls recommended by the contractor, 30 minutes shutdown of the unit, and 15 minutes for the unit to stabilize prior to another cycle. At every cycle the air conditioner shall be titled at a different position, 90 degrees apart, at an angle of 8 1/2 degrees to the true horizontal plane for satisfactory operation at all components during tilt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.6.11.4 High altitude hot day test 50 hours. The 50 hours shall be conducted in cycle of five hours duration. Each cycle shall consist of the following: Four hours of continuous operation, 15 minutes of recommended safety controls, 30 minutes shutdown of the unit and 15 minutes of warmup for the unit to stabilize prior to another cycle. At every cycle, the unit shall be tilted at a different position, 90 degrees apart, at an angle of 8 1/2 degrees to the true horizontal plane for satisfactory operation of all components during tilt. This test may be conducted in an altitude chamber. The condenser air mass flow shall be adjusted to simulate the flow rate at 5000 feet altitude.

* 4.6.12 Performance, low ambient and high altitude tests (200 hrs). The 200 hours performance and low ambient temperature test shall be conducted under the conditions specified in Table I with the unit wired for 230/60Hz operation and than 460 Hz operation. During the performance test the unit shall operate for 50 hours at 230V/60Hz and 50 hours at 460/60Hz. During the low ambient temperature test the unit shall operate for 25 hours at 230V/60Hz and 25 hours 460V/60Hz. It shall be demonstrated that the instantanous current does not exceed those currently specified in 3.16.1.

TABLE I Performance, low ambient and high altitude tests.

Hour of Operations	Ambient conditions	Leaving conditions	Air to be conditioned
*25	Rated capacity @ 100° F ±3° F DB/85° F ±3° F WB	45 P/M @ 50° F ±3° F against static pressure of 3 psig	Fresh air
25	Design summer day 100° F ±3° F DB/76° F ±3° F WB	60 P/M @ 50° F ±3° F against static pressure of 3 psig	Fresh air
*25	Design hot day 125° F ±3° F DB/75° F ±3° F WB	50 P/M @ 50° F ±3° F against static pressure of 3 psig	Fresh air
25	Design humid day 97° F ±3° F DB/87° F ±3° F WB	45 P/M @ 50° F ±3° F against static pressure of 3 psig	Fresh air
50	0° Ambient	45 P/M @ 50° F ±3° F against static pressure of 3 psig	Fresh air
50	High altitude 107° F ±3° F DB/87° F ±3° F WB (12.23 psia maximum barometric pressure)	45 P/M @ 50° F ±3° F against static pressure of 3 psig	Fresh air

Source: https://assist.dla.mil - Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.6.12.1 Data. At intervals of 30 minutes,(items specified in paragraph 4.6.11) sufficient data shall be recorded to determine compliance of the air conditioner with the performance requirements specified in Section 3. However, during the last half of the time specified in Table I for each design ambient condition data shall be recorded more frequently to show the characteristics of the temperature control until stability is reached, the length of time required to reach stability, and the ability of the control to maintain stability. These unstable conditions shall be created by varying either the airflow from maximum to minimum or varying the discharge pressure from 0 to 3 psig, or a combination of both. The order of tests in Table I shall be at the option of the contractor but the total continuous hours shall amount to those specified. One run or condition shall be not less than one hour duration.

4.6.12.2 200 Hour operation. The 200 hour portion as specified in Table I shall be so conducted as to provide sufficient data for determining the complete performance characteristic of the air conditioner of the specified ambient operating range.

4.6.12.3 Performance check. A performance check shall be conducted on the air conditioner under Design Summer Day (see 3.6.1.4) and within 10 minutes after the prime mover starts, the air conditioner shall be capable of delivering 60 pounds per minute of 50° F ±3° F at 3 psig. This shall be accomplished at intervals of 100 hours of operation and at end of 600 hours specified in para 4.6.11. The air conditioner shall operate for five hours under the Design Summer Day conditions. Upon conclusion of the 600 hour test the air conditioner shall demonstrate for five hours at each condition that it will meet the requirements of 3.6.1.2 through 3.6.1.8. The performance check shall be of sufficient duration to record all necessary data to ascertain satisfactory operation but shall not be in excess of five hours. This time may be logged against the endurance test.

* 4.6.13 Radio interference. The complete air conditioner shall be tested for radiated radio interference suppression in accordance with MIL-STD-462. The power cable shall be tested for conduction radio interference in accordance with MIL-STD-462.

* 4.6.14 Acoustic noise level test.

4.6.14.1 Test environment.

4.6.14.1.1 Test site. The test site for the air conditioner shall be on a concrete surface of uniform grade, free from gravel and other loose material. The air conditioner shall not be within 30 feet of a reflecting surface such as a aircraft hangar, building, tree or hillside.

4.6.14.1.2 Background noise. The noise level including wind noise shall be at least 10 db below that of the equipment noise being measured; but shall always be at least 10 db below the criteria. A windscreen shall be used for all measurements and the measurements shall not be made at velocities of 12 mph or over.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

4.6.14.1.3 Acoustic noise level testing. The air conditioner shall be tested for acoustic noise level while the unit is operating at maximum cooling capacity consistent with the prevailing ambient temperature. The criteria specified in 3.5.8 gives the noise levels at 10 feet. The nose levels at 10 feet shall be determined from measurement at head level (5 feet -4 inches) on a perimeter 10 feet from the exterior surface of the unit. The noise level for each octave band shall be determined at 12 position, 30 degrees apart around the perimeter. At no measurement position (10 ft) shall the values measured exceed the criteria limits specified in 3.5.8. The 60 foot flexible duct shall be attached to the outlet of the air conditioner and laid out to discharge the air away from the air conditioner. Sound level measurements shall also be taken at the operator's position.

4.6.14.1.4 Sound level measuring instrumentation. The audio sound level measurement shall be taken using a sound meter in conjunction with an octave band analyzer. The instruments used shall meet the requirements specified in ANSI Standard S1.4 (Latest Edition) except that the octave band analyzer shall employ the preferred frequencies specified in ANSI Standard S1.6 (Latest Edition). The flat frequency repsonse (dba and dbc readings will also be made at each measurement position including the instrument control panel) of the sound level meter in making all measurements. The instrumentation shall have a frequency response range of at least 45 to 11, 200 Hz, with the response known with +2db. An acoustical calibrator shall be used for an end-to-end (from the microphone to the meter) system calibration to provide the information needed for correcting the measured levels. The sound pressure level [in db (0,0002 dynes/CM^2)] produced by the calibrator when used in conjunction with the particular sound level pickup employed in the system shall be known to within $\pm$0.5 db.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements as Section 5 or the documents specified herein.

5. PACKAGING

5.1 Preservation, packaging and packing (See 6.2)

5.1.1 Level A and C. The air conditioner unit shall be preserved in accordance with MIL-R-3593,

or

Commercial Packaging in accordance with ASTM D 3951.

5.1.2 Levels A and B. The air conditioner, preserved and packaged as specified in 5.1, shall be shipped as a mobile unit; a shroud conforming to MS27766 shall be fabricated and secured over the unit.

5.1.3 Level C. Level C requirements shall be the same as Levels A and B except that the shroud will be omitted.

5.2 Packing. (See 6.2)

5.2.1 Level A, B, and C. The trailer mounted air-conditioner unit, Component parts, and repair parts shall be packed in accordance with MIL-STD-281, mobile.

or

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

Commercial packing in accordance with ASTM D 3951.

5.3 Markings. In addition to any special marking required (see 6.2) packages and exterior shipping containers, and unboxed (Mobile) units shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The A/M32C-4 air conditioner covered by this specification is intended for general purpose air conditioning of aircraft avionics during maintenance for preflight periods, maintenance shelters, portable hangars, and other similar enclosures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.

b. Level of preservation and packaging and level of packing required (see 5.1). Level B, preservation and packaging may be specified under the conditions of MIL-R-3593.

6.3 Definition.

6.3.1 Dry air conditioners. Dry air conditioner means that all fuels have been completely drained; however, this does not include all oil or refrigerant drainage.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 International standardization agreement. Certain provisions of this specification are the subject of International Standardization Agreement ABC Air Std 17/33. When amendment, revision or cancellation of this specification is proposed, which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

Review activity:
DLA - GS

Project 4120-F292

FSC 4120

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

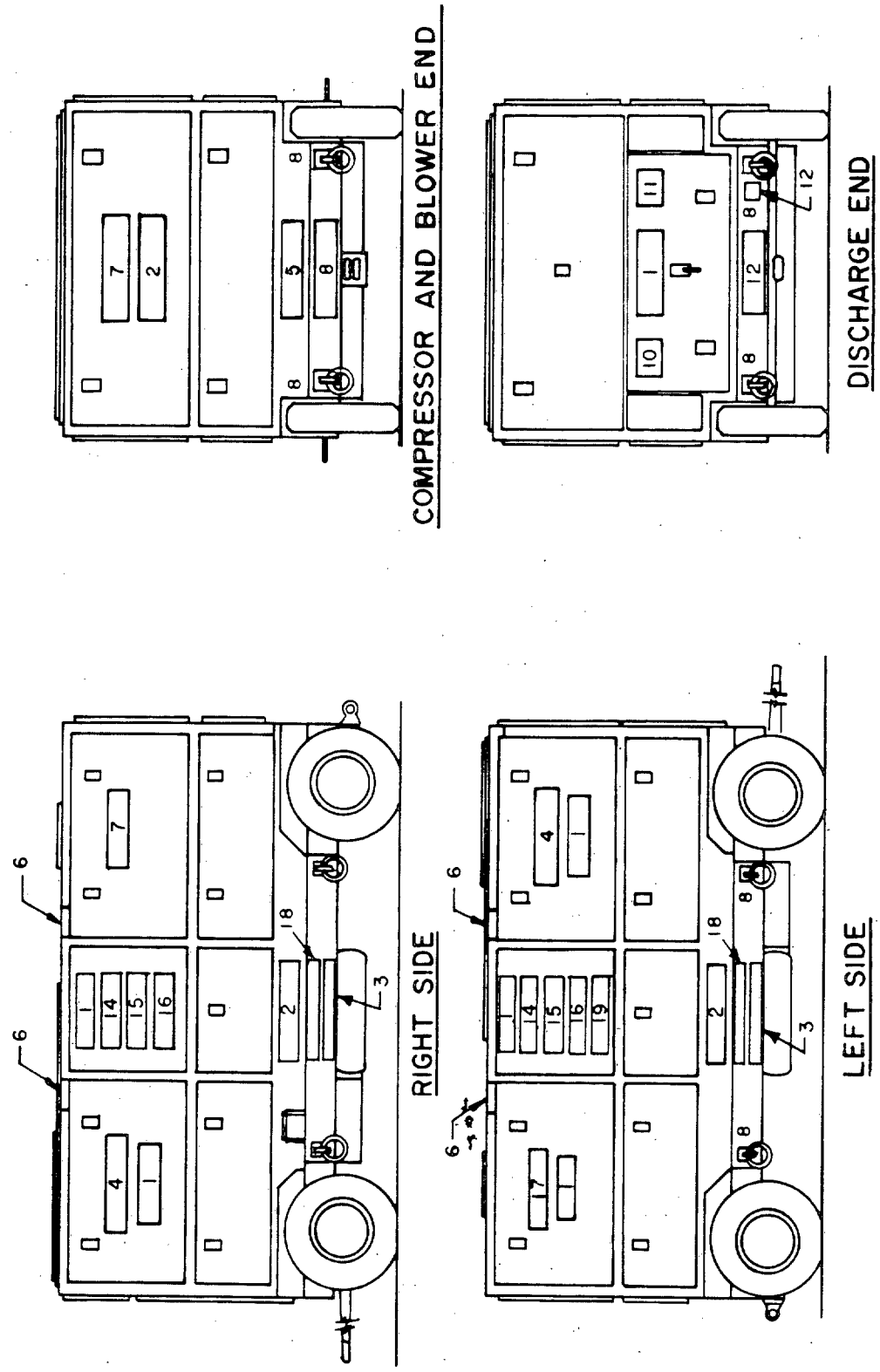


FIGURE 1. A/M32C-4 TRAILER MOUNTED AIR CONDITIONER

Source: https://assist.dla.mil - Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

A/M32C-4 AIR CONDITIONER

1) OPEN DOOR BEFORE OPERATING AIR CONDITIONER

2) CABLE GUIDE TO BE USED WHEN HOISTING

3) TIRE PRESSURE 100 PSIG

4) OPEN CONDENSER DOOR ON TOP BEFORE OPERATING AIR CONDITIONER

5) DO NOT TOW MORE THAN ONE SIMILAR UNIT FROM THIS PINTLE HOOK

6) CABLE GUIDES

7) ENGINE SERVICE DOOR

8) HOISTING/TIEDOWN RINGS

9) BEFORE TOWING, INSTALL PINTLE PIN AND INSERT SAFETY PIN

10) AIR INLET (INSIDE DOOR)

11) AIR OUTLET (INSIDE DOOR)

STENCILS TO BE PUT ON ENCLOSURE & VISABLE WITH DOOR OPEN

12) HANDBRAKE

13) MAXIMUM TOWING SPEED 20 MPH

14) US AIR FORCE

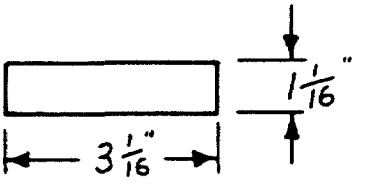
15) REGISTRATION NUMBER - LEAVE BLANK

16) A/M32C-4 AIR CONDITIONER

17) CONTROL PANEL

18) CENTER OF BALANCE MARKING

19) BLOWER SERVICE DOOR



NOTES: 1. ALL LETTERS TO BE 1" HIGH EXCEPT ITEMS 14, 15, 16, ITEMS 9, 18 TO BE 1/2" HIGH.

2. CENTER STENCIL ON UNIT AS SHOWN UNLESS OTHERWISE SPECIFIED.

3. ALL STENCILS TO BE BLACK CONFORMING TO FED-STD-595, COLOR 27038 OR 37038 EXCEPT ITEM 9 TO BE YELLOW CONFORMING TO FED-STD-595, COLOR 13538.

FIGURE 1. A/M32C-4 AIR CONDITIONER. - CONTINUED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

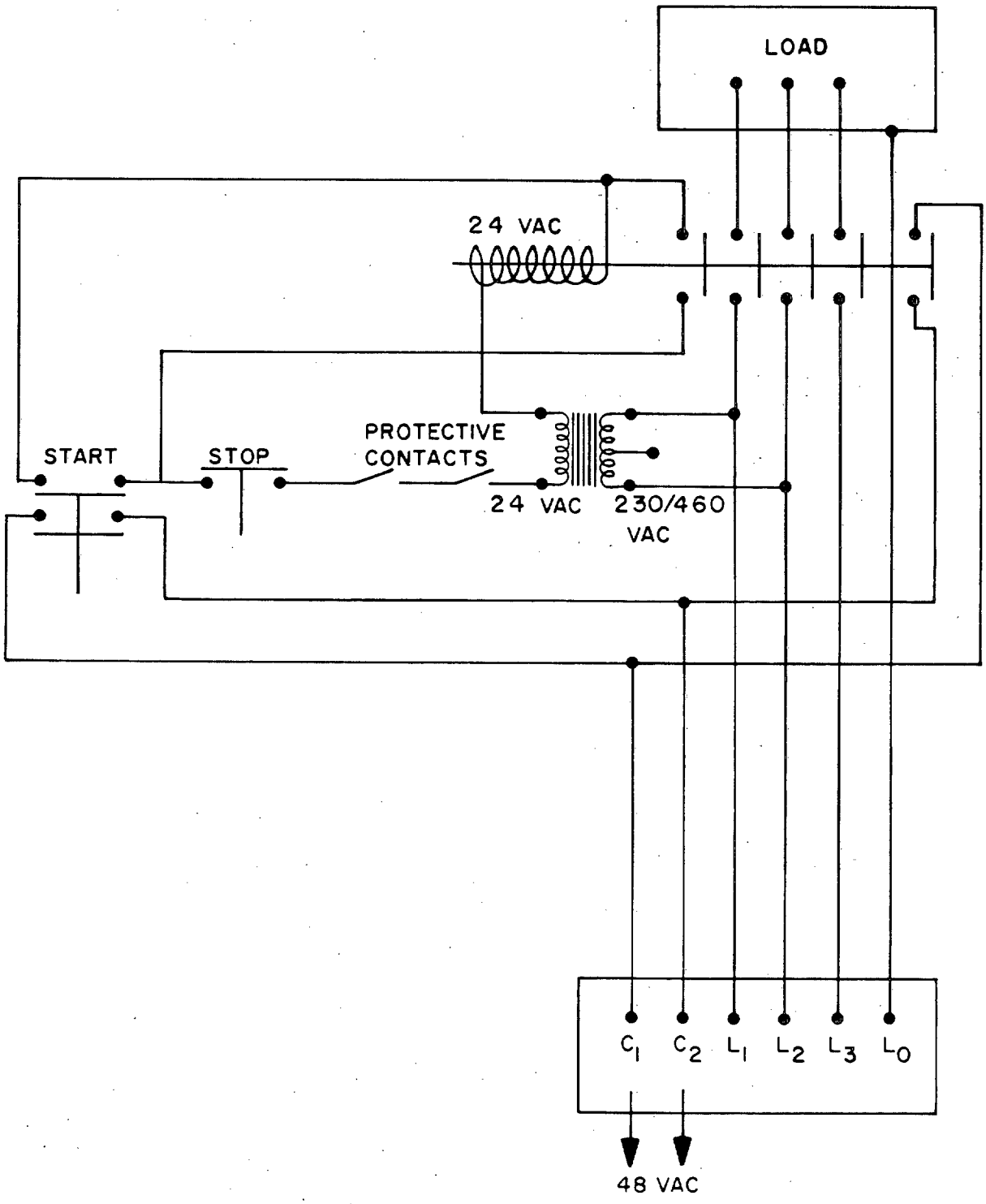


Figure 2. STARTING CONTROL CIRCUIT WIRING DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASH DC

POSTAGE WILL BE PAID BY HQ U.S. AIR FORCE

SA-ALC/MMEDO
KELLY AFB, TX 78241

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-26846B	AIR CONDITIONER, A/M32C-4, TRAILER MOUNTED, VAPOR CYCLE, ELECTRIC MOTOR DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:25Z
Check the source to verify that this is the current version before use.